

[**N.B.** – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **THREE** questions taking at least ONE question from each group]

Group - A

1. $x_1 = 5, x_2 = 10, x_3 = 12, x_4 = 14, x_5 = 20$

- (a) What is a sample?

1
- (b) Find $\sum_{i=1}^5 (x_i - 5)$

2
- (c) Determine $\sum_{i=1}^5 3(x_i - c)^2$, where c is the last digit of your cadet number.

3
- (d) Find mean of square and square of mean from the data. When will they be equal?

Group - B